

A Compact Dual Band Octagonal Shaped Patch Antenna for Ku-Band and 5G Applications

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Abstract—This study presents a slotted dual-band octagon-shaped patch antenna feed by a microstrip line, designed for Ku-band and upper 5G (mmWave) applications. The -10 dB bandwidth of the suggested antenna is 3.412 GHz in the first band and 6.055 GHz in the second operational region, which accommodates substantial data volume. The antenna operates inside the frequency range of 12.14 GHz to 15.552 GHz for Ku-band applications and 25.061 GHz to 31.116 GHz for 5G usage. The octagonal slotted antenna with a partial ground plane resonates at 13.954 GHz and 28.132 GHz, exhibiting return losses of -21.51 dB and -45.02 dB, respectively. Rogers RT-5880 serves as the substrate material dimensions of $20 \times 20 \times 0.79$ mm³ provides a peak gain ranging from 4.08 dB and 7.03 dB, maintains a VSWR below 2 throughout the operating bands, and delivers an average radiation efficiency of 95%. These characteristics make the proposed antenna a promising candidate for high-speed satellite and next generation 5G communication systems.

Keywords—Patch antenna, dual band, octagon slot, 5G, mmWave, Ku band, satellite communication.

I. INTRODUCTION

The rapid advancement of wireless technologies, particularly 5G mmWave (24 GHz and above) and Ku-band satellite systems (12–18 GHz), has increased the requirement for tiny, extremely effective antennas that can operate across several bands. mmWave frequencies enable ultra-fast data rates (up to 10 Gbps), low latency, and high capacity for applications such as AR, cloud gaming, and UHD streaming, while the Ku band supports satellite internet, broadcasting, and VSAT services with reduced atmospheric loss. Microstrip patch antennas, with their low profile, ease of fabrication, and planar integration, are promising candidates for efficient terrestrial and satellite communication solutions. [1]. The progression of technology requires a multiband frequency and broadband compact antenna to improve the efficacy of wireless communication systems [2]. Dual-band microstrip patch antennas are vital for modern wireless systems, offering compact size, low cost, and reliable broadband performance. Its lightweight, surface-mountable design with simple feed

integration makes them highly suitable for integrated terrestrial and satellite communication applications [3]. Thus, these are suitable for various applications, including mobile phones, satellites, GPS, and radar systems [4]. The rapid growth of wireless communication demands compact, low-cost antennas with high gain, wide bandwidth, low loss, and over 70% efficiency to support increased capacity, elevated data rates, and reliable performance [5]. 5G wireless technology signifies a potential solution for multi-gigabit per second data rates in forthcoming mobile communications. 5G wireless technology represents a possible option for multi-Gbps data rates in future mobile communications [6], [7].

A microstrip patch antenna in [8] proposed for Ku-band usage attained a return loss of -26.55 dB and a gain of 6.31 dB, and excellent radiation efficiency. A slotted microstrip patch antenna (MPA) for Ku-band with a imperfect ground is reported in [9], with a volume of 3052.8 mm³, dual centre frequencies, apex gain of 3.67 dB, and VSWR < 2. Moreover L. C. Paul et al. [10] developed a wide-band MPA for 5G resonated at 27.47 GHz, providing 5.22 dB gain, 70% efficiency, and low VSWR. A rectangular MPA with partial ground for 5G is presented in [11], covering 2.567 GHz bandwidth, -31.67 dB return loss, 535.5 mm³ volume, and 8.39 dB gain. In [12], a low-profile MPA with DGS for 27.26 GHz yielded 5.11 dB gain but lower efficiency. A wide-band MPA with modified ground and double T-shape is introduced in [13], achieving 7.468 GHz bandwidth, 6.367 dB gain, and 92% efficiency. In [14], a slotted MPA for 28 GHz showed 8.104 dB gain, 89.77% efficiency, and low VSWR. However, many dual-band antennas remain relatively large, limiting integration into compact 5G payloads. Attaining substantial gain and extensive bandwidth in a miniaturized configuration remains challenging, with many designs suffering from degraded radiation performance. Among these, dual-band antennas have attracted considerable attention to address the increasing demand for multi-standard devices, particularly for applications spanning 5G millimetre-wave (mmWave) communications and Ku bands (12–18 GHz) applications.

To address these gaps, this paper describes the design of a single-element, microstrip line-fed, dual-band, octagon-shaped patch antenna intended for Ku-band (12.141–15.552 GHz) and upper 5G (25.061–31.116 GHz) applications, employing the CST Studio Suite. The aim is to introduce a compact MPA with improved gain and radiation characteristics, offering an antenna solution that bridges the needs of terrestrial 5G and satellite Ku-band applications within a miniaturized and efficient structure.

II. DESIGN AND ANALYSIS

The design procedures, along with the descriptions of a double band octagon shape with a slot microstrip patch antenna with a partial ground plane, are discussed in this paragraph. The CST Studio Suite software is utilized to design and simulate the antenna. The Rogers RT 5880 (lossy) material of dielectric constant 2.2, a loss tangent of 0.0009 and thickness of 0.79 mm is used as the substrate of the designed antenna. In real time fabrication scenario, the copper layer (thickness 0.035 mm) is deposited on the substrate using photolithography process and the microstrip line is etched to a width of 2.45 mm to ensure a 50Ω impedance matching. The octagonal shape and the etched slot are then defined on the front side of the substrate, followed by the creation of the partial ground plane on the backside. After etching, the formed structure will be used to design as a main radiating element. The total volume of the antenna is $20 \times 20 \times 0.79 \text{ mm}^3$. At Stage-1 in Fig. 1, a rectangular-shaped patch is constructed using annealed copper on the top of the substrate, with a ground plane on the bottom of the substrate. Then a regular octagon shape of 5mm (all eight sides) is designed with a full ground plane contribute dual resonance point. To confirm decent impedance matching with the port impedance of 50Ω , the width of the microstrip line is taken as 2.45 mm, which is directly linked to the port of the MPA. In Stage-3, an irregular octagonal-shaped slot is etched into the radiating patch after that a ground plane is changed to a partial ground (Stage-4) while keeping the shape of the main radiating element unchanged. Finally, the partial ground plane is further modified to achieve a better reflection coefficient (Fig. 2a) and enhanced gain which is observed in gain vs frequency curve (Fig. 2b). Fig. 2 shows the performance of antenna in different design stages. The final stage exhibits a comparatively better reflection coefficient (-21.513 dB and -45.024 dB) than the previous stages at dual resonance frequency of 13.954 GHz and 28.132 GHz respectively. Optimized dimensions of the proposed MPA are tabulated in Table I. The front sight and the back sight of the designed antenna are presented in Fig. 3(a) and Fig. 3(b), whereas the 3D view of the designed antenna is shown in Fig. 3(c).

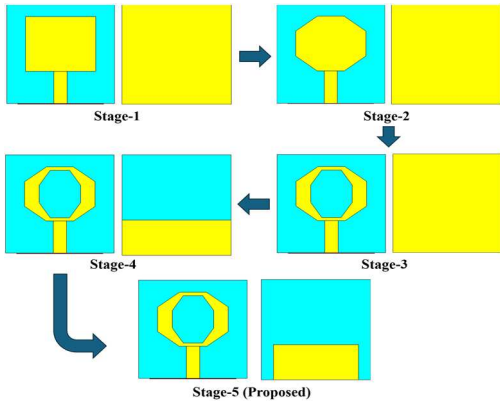


Fig. 1. Design evaluation of proposed dual band octagon-shaped MPA.

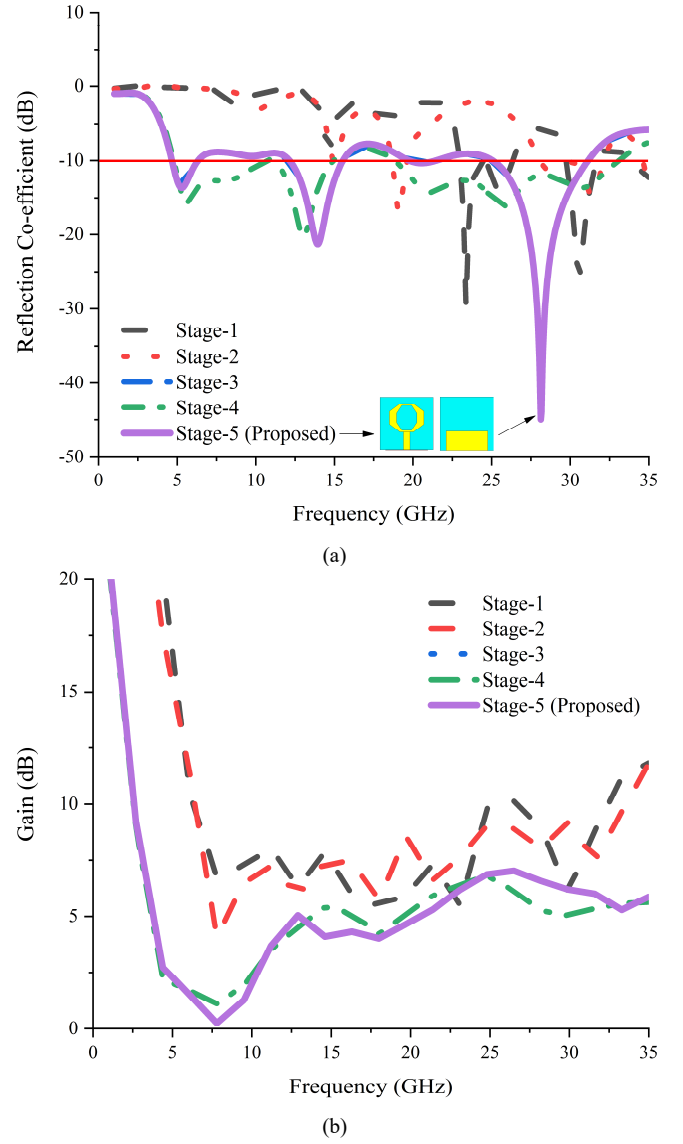


Fig. 2. Performance evaluation in terms of (a) reflection co-efficient and (b) gain analysis of the proposed octagon-shaped MPA.

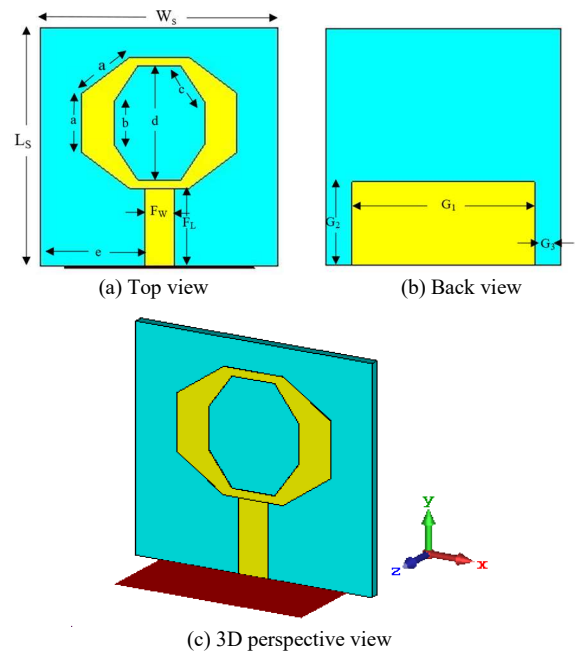


Fig. 3. Proposed octagon-shaped dual-band MPA.

TABLE I. DESIGN SUMMARY OF OCTAGON-SHAPED DUAL-BAND MPA

Antenna Parameters	Length (mm)
Substrate width, W_s	20.00
Substrate Length, L_s	20.00
Outer octagonal arms, a	5.00
Inner rectangular arm length, b	3.60
Inner octagonal arms, c	3.61
Inner slot height, d	9.60
Distance of feed, e	8.77
Feeder width, F_w	2.45
Feeder length, F_L	6.50
Partial ground plane width, G_1	15.50
Partial ground plane length, G_2	7.00
Gap of ground plane, G_3	2.25

III. PERFORMANCE ANALYSIS

A. Reflection Co-efficient

The performance analysis of the double-band octagon-shaped microstrip patch antenna is carried out by evaluating vital parameters such as return loss, bandwidth, radiation pattern, and radiation efficiency. The return loss is examined to ensure decent impedance matching and minimal signal reflection at the designed frequency bands. The bandwidth of the antenna is measured to determine its capability to operate effectively over an extensive range of frequencies. The curve that indicates the return loss of the octagon-shaped compact double-band microstrip patch antenna with partial ground plane is demonstrated in Fig. 4. The center frequencies of the proposed antenna are 13.954 GHz and 28.132 GHz, with the values of return loss of -21.51 dB and -45.02 dB, respectively. The first operating zone is fit for Ku band applications and the second operating region is suitable for 5G applications. The bandwidths (considered at -10 dB points) of the designed antenna are 3.412 GHz for the first operating region and 6.05 GHz for the 5G applications. An excellent bandwidth coverage for the upper 5G and a moderate bandwidth coverage for the Ku band applications are indicated here.

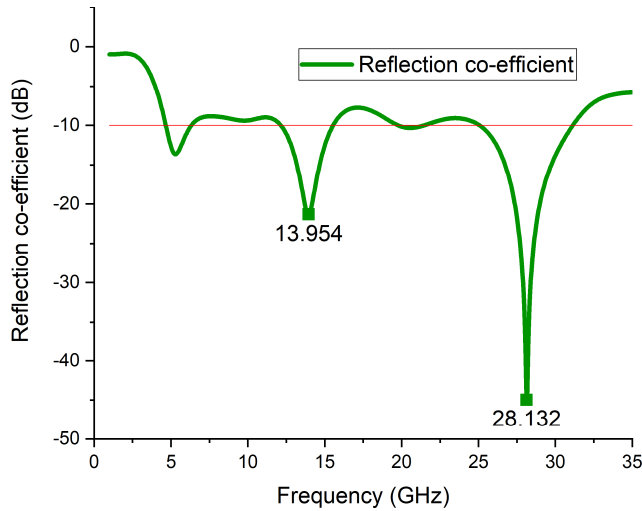
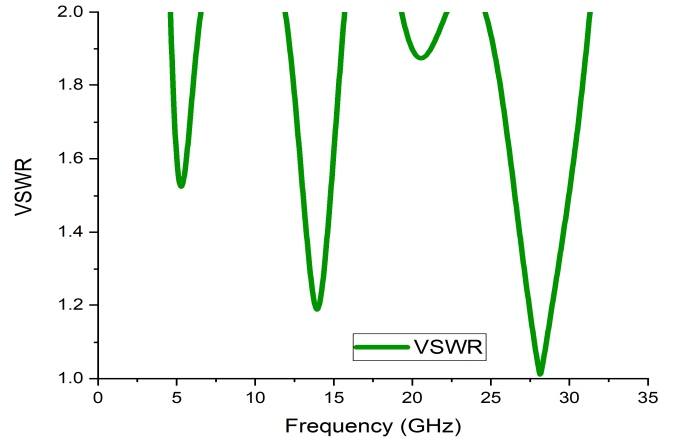
Fig. 4. $|S_{11}|$ curve of the double-band octagon-shaped MPA.

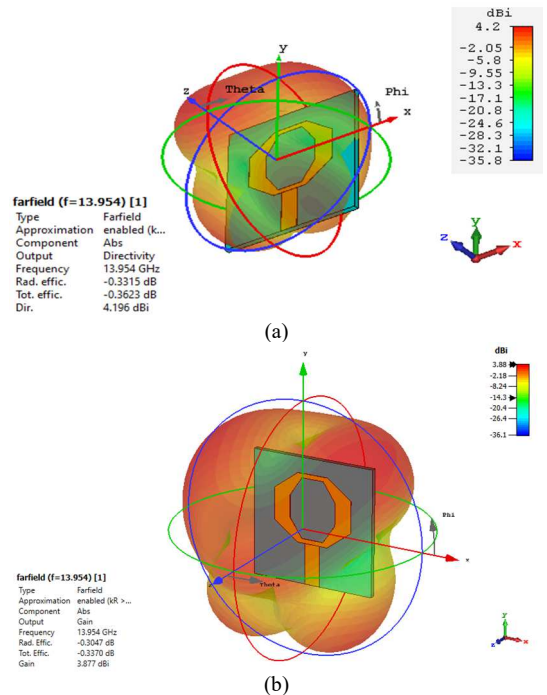
Fig. 5. VSWR of the double-band octagon-shaped MPA.

B. Voltage Standing Wave Ratio

VSWR characterizes the quality of impedance matching. The low value close to one of VSWR indicates efficient transmission and better performance. The VSWR over the complete operating region is about $1 \leq \text{VSWR} \leq 2$. In the first operating region, the values range from approximately 1.18 to 1.92, whereas in the second operating region, they range from 1.01 to 1.92. The Fig. 5 shows the fluctuation of VSWR over both the operating region. The VSWR at the first resonant point is 1.18 and second the resonant point is 1.01. So, It is nearly equal to unity for the second band. The impedance matching is almost excellent in the case of this designed prototype.

C. Radiation Pattern

Gain and Directivity both are important criteria for assessing an antenna's efficacy. Gain specifies how efficiently an antenna translates input power into radio waves in a precise direction whereas directivity indicates antenna's energy focusing in a specific direction compare to an isotropic antenna. The 3D radiation pattern of the first and second operating band of the designed double-band octagon-shaped MPA are demonstrated in Fig. 6(a, b) and Fig. 6(c, d) respectively in terms of directivity and gain.



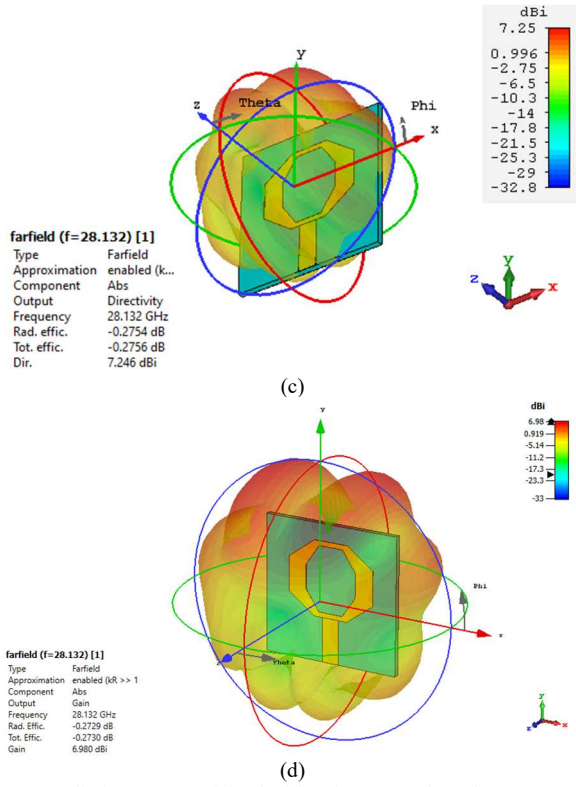


Fig. 6. Radiation pattern of the dual-band octagon-shaped MPA at (a, b) 13.954 GHz and (c, d) 28.132 GHz.

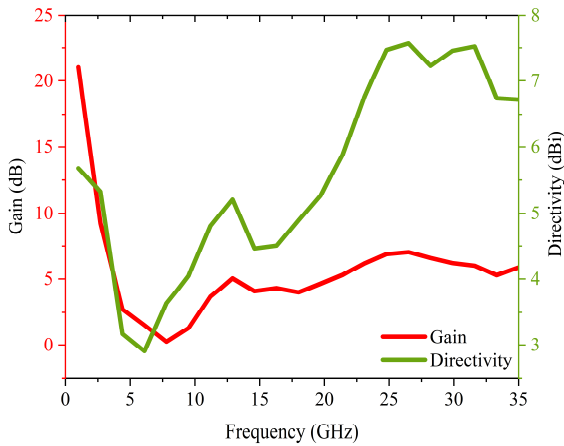


Fig. 7. Gain and directivity vs frequency curve of the double-band octagon-shaped MPA.

Fig. 7 illustrates the gain and directivity versus frequency characteristics of the double-band octagonal MPA. At the first resonance frequency, the antenna proclaims a peak gain of 4.44 dB and a directivity of 4.72 dBi, with the gain and directivity ranging from 4.08–5.05 dB and 4.45–5.17 dBi, respectively, across the first operating band. At the second resonance frequency, a maximum gain of 6.59 dB and a directivity of 7.25 dBi are achieved, while the gain and directivity span 6.20–7.03 dB and 7.21–7.57 dBi, respectively, over the second operating band.

The polar plot of radiation pattern of electric field (E-field) and magnetic field (H-field) at the azimuth angle $\varphi=0^\circ$ and $\varphi=90^\circ$ for the initial operating region are depicted in Fig. 8(a) at resonant frequency 13.954 GHz. From the polar plot of E-field, it is evident that at $\varphi=90^\circ$ the lowest side lobe level is -8.7dB and peak half power beam width is 44.7° and the primary lobe magnitude is 18.6 dBV/m whereas at $\varphi=0^\circ$, the primary lobe magnitude is 17.6 dBV/m. The primary lobe

magnitudes are -33.9 dBA/m at $\varphi=0^\circ$ and -32.9 dBA/m at $\varphi=90^\circ$ observed from the polar plot pattern related to H-field. For $\varphi=90^\circ$, the primary lobe is directed at 13° in the case of H-field. The polar scheme of radiation pattern of electric field (E-field) and magnetic field (H-field) at the azimuth angle $\varphi=0^\circ$ and $\varphi=90^\circ$ for the second operating zone are depicted in Fig. 6(b) at resonant frequency 28.132 GHz. From the polar plot of E-field, it is clear that at $\varphi=90^\circ$ the lowest side lobe level is -4.7dB and maximum half power beam width is 53.6° and the main lobe magnitude is 16.1 dBV/m whereas at $\varphi=0^\circ$, the main lobe magnitude is 13.2 dBV/m. The primary lobe magnitudes are -38.4 dBA/m at $\varphi=0^\circ$ and -35.4 dBA/m at $\varphi=90^\circ$ observed from the polar plot pattern related to H-field. For $\varphi=90^\circ$, the primary lobe is directed at 37° and for $\varphi=0^\circ$ the primary lobe is directed at 42° in the instance of H-field.

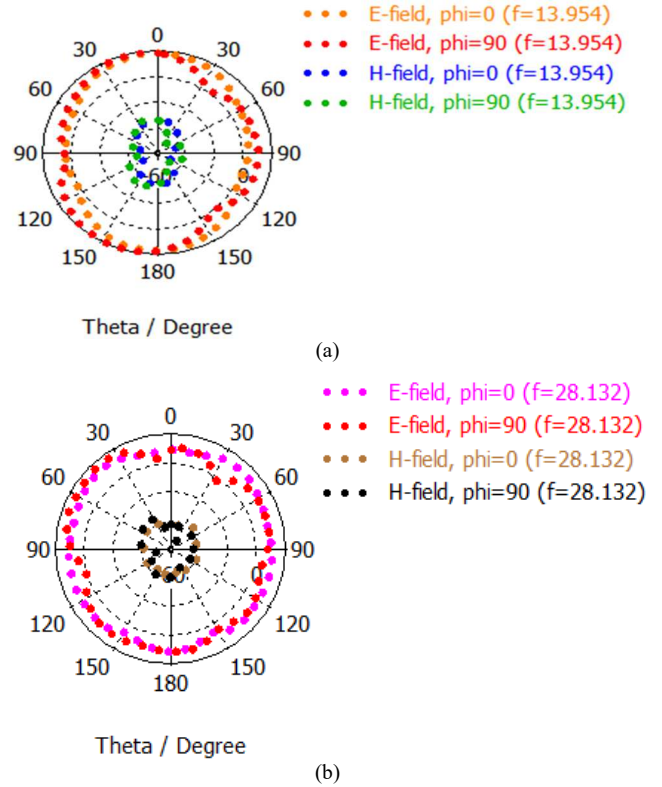


Fig. 8. E, H-field polar pattern at (a) 13.954 GHz; (b) 28.132 GHz

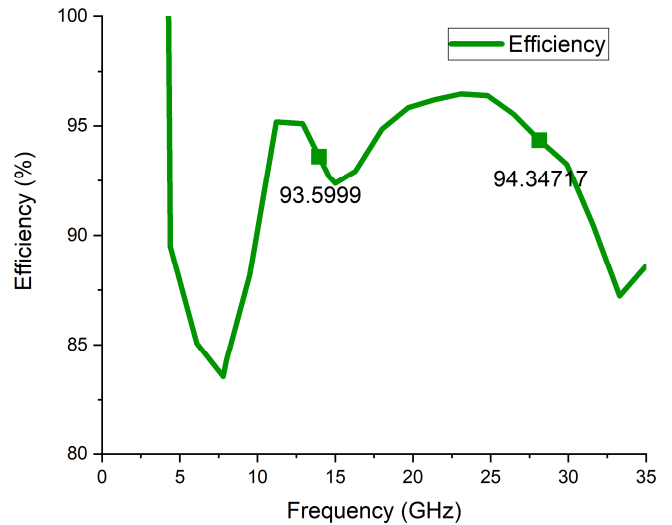


Fig. 9. Radiation efficiency of the proposed MPA

D. Radiation Efficiency

Radiation efficiency is an evaluation of the capability of an antenna to translate the electrical power that is provided to it into electromagnetic radiation that spreads through free space. The graph of radiation efficiency vs operating frequency of the proposed dual-band octagon-shaped MPA is shown in Fig. 9. It is observed that in the initial operating region the radiation efficiency varies from 92% to 95%. In the other operating region it varies from 93% to 95%. So, the proposed antenna is capable to convert most of the electrical power it receives into electromagnetic waves for the propagation through the free space.

E. Surface Current Distribution

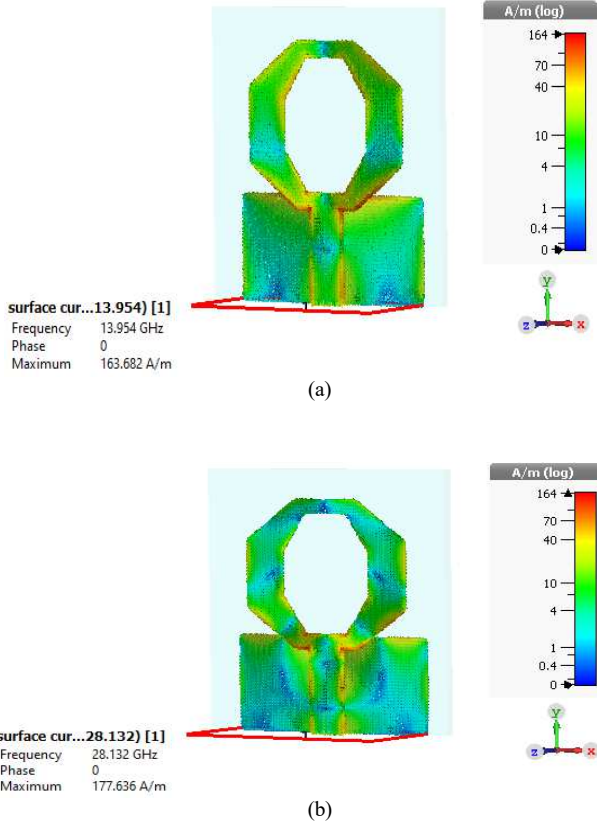


Fig. 10. Surface current of the proposed MPA at (a) 13.954 GHz and (b) 28.132 GHz.

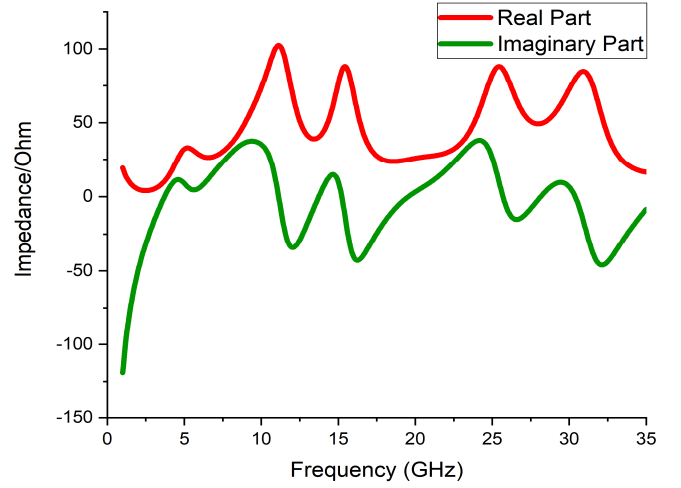


Fig. 11. Impedance analysis of the proposed octagon-shaped MPA.

The simulation result of surface current distribution for the proposed dual-band MPA at both 13.954 GHz and 28.132 GHz are shown in Fig. 10. The figures demonstrate that the current concentrations are higher at the edges. The maximum current density at the first operating band is 163.68 A/m whereas in the second operating band the value is 177.63 A/m. The impedance both the real and imaginary portion of the designed prototype is shown in Fig. 11 for both the operating regions. The MPA possesses impedance of $(42.78 + j4.84) \Omega$ at the first center frequency of 13.954 GHz and $(49.321 - j0.075) \Omega$ at the later center frequency of 28.132 GHz.

IV. COMPARATIVE ANALYSIS

Compared to prior works in TABLE II, the proposed miniaturized octagonal slotted dual-band patch antenna ($20 \times 20 \times 0.79 \text{ mm}^3$) covers both Ku band and 5G mmWave frequencies. It achieves bandwidths of 3.412 GHz and 6.055 GHz, excellent VSWR (~ 1.01 – 1.18), and $\sim 95\%$ radiation efficiency. This compact design surpasses or matches larger, complex antennas, overcoming earlier limitations of bandwidth, size, and single-band operation for next-generation wireless and satellite systems.

V. FUTURE WORK

Future work explores metamaterials and flexible substrates to improve bandwidth, efficiency, and resilience of microstrip patch antennas. Integration of beamforming and MIMO will enhance signal strength and capacity for 5G and satellite networks. Cost-effective fabrication via additive manufacturing and 3D printing ensures precision, scalability, and reliable next-generation communication.

TABLE II. COMPARISON WITH RECENT RELEVANT WORKS

Reference	Dimensions (L×W×h) mm ³	Operating frequency range (GHz)	BW (GHz)	Centre frequency (GHz)	Reflection Co-efficient (dB)	Max. Gain (dB)	Radiation Efficiency (%)	VSWR	Application
[8]	14.1×18.6×1.6	11.595 – 15.75	4.155	12.54	-26.55	6.31	98.99	1.0986	Ku Band
[9]	40×48×1.59	12.80 – 15.83	3.03	12.8, 14.9	-25, -17	3.67	—	< 2	Ku Band
[10]	20×20×0.79	26.154 – 31	4.846	27.47	-35.03	5.22	70	1.036	5G
[11]	20×17×1.575	26.984 – 29.551	2.567	28.201	-31.677	8.39	84.5	< 1.05	5G
[12]	20×20×0.79	25.52 – 31.37	5.85	27.26	-35.353	5.11	66	1.034	5G
[13]	30×30×1.575	24.333 – 31.801	7.468	26.88	-40.36	6.367	92	1.0194	5G
[14]	40×30×1.575	24.356 – 29.197	4.841	25.69	-42.134	8.104	89.77	1.0158	5G
This work	20×20×0.79	12.14 – 15.552 25.061 – 31.116	3.412 6.055	13.954 28.132	-21.513 -45.024	4.44, 6.59	95	1.18 1.01	Ku Band, 5G

VI. CONCLUSION

The growing demand for microstrip patch antennas for 5G mmWave and Ku-band satellite communication is driven by expanding 5G networks, increased mobile data consumption, and the need for global connectivity. 5G mmWave provides exceptionally high data rates, minimal latency, and substantial capacity, while Ku-band satellites provide broadband, VSAT, and broadcasting services, especially in remote areas. The proposed dual-band MPA with a partial ground plane effectively supports both bands, attaining return losses of -21.513 dB and -45.024 dB, maximum gains of 5.05 dB and 7.24 dB, VSWR between 1 and 2, and $\sim 95\%$ radiation efficiency. Its compactness, simple structure ensures optimal impedance matching and high signal strength. Integrating 5G mmWave with Ku-band satellite communication enhances global connectivity, enabling 5G satellite backhaul and reliable high-speed communication. Performance and versatility of proposed antenna make it an ideal solution for next-generation terrestrial and satellite wireless systems.

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